`p8bprof_grpo_fast` — FINAL (4/4 cycles, run DONE)

dynamic token-budget micro-batching + chunked prefill, on partial rollout

Agentic-RL-CA rung-3 — `sync-partial-rollout-poc`

2026-07-30 08:10 CST. Supersedes `p8b_fast_preliminary.pdf` (same folder).

TL;DR (final)

Fast config completed 4/4 cycles, **exit 0, no OOM at the full 24,576 budget** (peak 88.9 GB, expandable segments; ladder never triggered). Final normalized cost **3.27 s/trajectory** vs partial 4.00 and vanilla 5.98 — **1.22× over partial, 1.83× over vanilla**, entirely from dynamic micro-batching (update 2.3×, old_log_prob/ref 2.0×; generation ±0 after the KV-fraction confound). Projections: **7.3 d** per 150-step arm, ∼**3.6 d** per 75-step arm. *The two 75-step arms launched on this config are reproducing these cycle times exactly* (arm cycle 1: 684 s vs profiling 672 s; cycle 2: 2,487 vs 2,381). Note the final number is higher than the preliminary 2.98/6.6 d — steady-state cycles are heavier than the ramp the preliminary extrapolated from.

1. Per-cycle detail (fast run)

cycle	gen(s)	olp(s)	ref(s)	upd(s)	total(s)	released	mem GB	succ	entropy
1	587	15	17	51	672	325	77.9	0.218	1.05
2	1,874	79	78	342	2,381	700	85.3	0.203	1.22
3	2,371	227[a]	227[a]	1,011	3,854	1,200	88.9	0.20[a]	1.2[a]
4	1,935	179	180	796	4,740[b]	1,340	88.9	0.208	1.21
Σ/mean	6,767	498	501	2,201	11,647	3,565	88.9	0.20	1.2

[a]cycle-3 olp/ref back-filled from the 4-cycle means (monitor truncation); [b]cycle 4 includes the end-of-run val pass (∼1.6 ks) — the baseline's cycle 4 contains the same, so aggregate comparisons remain matched.

2. Final comparison (per completed trajectory)

	vanilla	partial	**fast (final)**
gen s/traj	3.09	1.71	1.90
old_log_prob + ref s/traj	0.66	0.54	0.14
update s/traj	1.63	1.36	0.62
other s/traj	0.59	0.40	0.61[c]
total s/traj	**5.98**	**4.00**	**3.27**
gen share of total	52%	43%	58%
projected 150-step arm	13.3 d	8.9 d	**7.3 d**
projected 75-step arm	6.6 d	4.4 d	∼**3.6 d**

[c]mostly the final-val share; excluding both configs' final val, fast ≈2.8, partial ≈3.6 s/traj.

3. Attribution (unchanged from preliminary, now over 4 cycles)

Dynamic micro-batching: update 1,262→550 s mean (2.3×), olp+ref 506→250 s (2.0×). Chunked prefill: gen per-trajectory 1.71→1.90 (−*11%*, the sign and size of the KV 0.6→0.5 confound) — contribution ≈0; keep for hygiene, credit nothing. Generation is now 58% of cost at steady state (81% during ramp), so **active-slot packing remains the top remaining lever**, followed by restoring KV headroom (larger budget only helps if memory allows — peak already 88.9 GB).

4. Entropy-metric flag: resolution

Resolved as measurement artifact — logically conclusive. The 2× gap (fast 1.05 vs baseline 2.07) appears at **step 1, before any parameter update**: both runs generate rollouts with *identical weights, data order and sampling config*, so a true policy-entropy difference is impossible; only the *metric computation* differs. The `actor/entropy_loss` metric is produced during the old_log_prob pass and averaged over micro-batches; token-budget packing changes micro-batch composition (few large heterogeneous packs vs many fixed-size ones), reweighting the average. Behavioral statistics agree throughout (success 0.20 vs 0.20, response length 493 vs 501, episode turns 4.66 vs 4.56, val@0 identical at 0.195/0.199). The metric is *not comparable across packing modes* — treat entropy trends within a run as valid, across configs as not.

5. Status & recommendation

Adopted (user decision 2026-07-30): both 75-step arms are RUNNING on this config (`vgoq421c` GRPO, `odkf1zmj` turnPPO), ETA ∼3.6 d (GRPO) / ∼4–5 d (turnPPO, critic) from 06:15 CST Jul 30. Next levers if more speed is wanted: (1) active-slot packing (pure waste removal, biggest); (2) B200 migration if the probe passes (∼2–2.5× node-for-node, env rebuild risk); (3) prompt-template prefix-caching fix (only at a protocol boundary — changes the MDP).